Python Libraries & Modules for AI/ML

Every important Python module for data, ML, DL, NLP, vision, deployment and LLMs: what it is, install command, and a working example.

Contents

#	Topic	Level
1	NumPy	Beginner
2	Pandas	Beginner
3	Polars	Intermediate
4	Matplotlib	Beginner
5	Seaborn	Beginner
6	Plotly and Dash	Intermediate
7	SciPy	Intermediate
8	Statsmodels	Intermediate
9	scikit-learn	Beginner
10	XGBoost / LightGBM / CatBoost	Intermediate
11	imbalanced-learn	Intermediate
12	Optuna / Hyperopt / Keras Tuner	Advanced
13	PyTorch	Intermediate
14	TensorFlow and Keras	Intermediate
15	JAX / Flax	Advanced
16	PyTorch Lightning / fastai / Hugging Face Accelerate	Advanced
17	ONNX and ONNX Runtime	Advanced
18	OpenCV	Intermediate
19	Pillow, scikit-image, Albumentations	Beginner
20	torchvision / timm / Ultralytics YOLO	Intermediate
21	NLTK	Beginner
22	spaCy	Intermediate
23	Gensim	Intermediate
24	Hugging Face Transformers / Datasets / Tokenizers / PEFT / TRL	Intermediate
25	sentence-transformers	Intermediate
26	LangChain / LlamaIndex / LangGraph	Intermediate
27	Anthropic / OpenAI SDKs	Beginner
28	Vector DBs: FAISS, Chroma, Qdrant, Pinecone, pgvector	Intermediate
29	Gymnasium and Stable-Baselines3	Advanced
30	Prophet / sktime / tsfresh / darts	Intermediate
31	SHAP / LIME / Captum	Advanced
32	NetworkX / PyTorch Geometric / DGL	Advanced
33	Surprise / implicit / LightFM / RecBole	Intermediate

#	Topic	Level
34	Streamlit / Gradio	Beginner
35	FastAPI / Flask / Uvicorn	Intermediate
36	MLflow / Weights & Biases / TensorBoard / DVC	Intermediate
37	Pydantic, python-dotenv, tqdm, joblib, rich	Beginner
38	Dask / PySpark / Ray / Modin	Advanced
39	Jupyter, ipywidgets, papermill, nbconvert	Beginner
40	Web scraping: BeautifulSoup, Scrapy, Selenium, Playwright	Intermediate
41	Speech & audio: librosa, whisper, torchaudio, SpeechRecognition	Intermediate

1. NumPy [Beginner]

Explanation (English)
N-dimensional arrays, broadcasting, vectorised maths, linear algebra, random numbers. Base of the whole ecosystem. Install: pip install numpy

Tanglish (Tamil + English) Vilakkam
NumPy fast array library. Broadcasting, vectorization. ML-oda base.

Example / Program
```
import numpy as np
a=np.arange(12).reshape(3,4)
print(a.mean(axis=0),a[a>5],a@a.T,np.random.default_rng(0).normal(size=3))
```

Learn more
YouTube search: NumPy python tutorial
Docs: https://numpy.org/doc/

2. Pandas [Beginner]

Explanation (English)
DataFrame/Series for loading, cleaning, merging, grouping and reshaping tabular data (CSV, Excel, SQL, JSON, Parquet). Install: pip install pandas

Tanglish (Tamil + English) Vilakkam
Pandas = Excel maadhiri tabular data handle panna. read_csv, groupby, merge.

Example / Program
```
import pandas as pd
df=pd.DataFrame({'a':[1,2,2],'b':[3,4,5]})
print(df.groupby('a').b.sum(),df.describe())
```

Learn more
YouTube search: Pandas python tutorial
Docs: https://pandas.pydata.org/docs/

3. Polars [Intermediate]

Explanation (English)
Fast DataFrame library written in Rust with lazy evaluation; an alternative to pandas for large data. Install: pip install polars

Tanglish (Tamil + English) Vilakkam
Polars pandas-a vida fast (Rust). Large data-ku nalladhu.

Example / Program
```
import polars as pl
df=pl.DataFrame({'a':[1,2,3]})
print(df.select(pl.col('a').sum()))
```

Learn more
YouTube search: Polars python tutorial
Docs: https://docs.pola.rs

4. Matplotlib [Beginner]

Explanation (English)
Core plotting library: line, scatter, bar, hist, subplots, saving figures. Install: pip install matplotlib

Tanglish (Tamil + English) Vilakkam
Matplotlib graph plot panna. line, bar, scatter, hist.

Example / Program
```
import matplotlib.pyplot as plt
plt.plot([1,2,3],[1,4,9]);plt.title('sq');plt.savefig('a.png')
```

Learn more
YouTube search: Matplotlib python tutorial
Docs: https://matplotlib.org/stable/

5. Seaborn [Beginner]

Explanation (English)

Statistical visualisation on top of matplotlib: heatmap, pairplot, boxplot, violin, regplot. Install: pip install seaborn

Tanglish (Tamil + English) Vilakkam

Seaborn azhagana statistical charts - heatmap, pairplot.

Example / Program

```
import seaborn as sns
df=sns.load_dataset('iris');sns.pairplot(df,hue='species')
```

Learn more

YouTube search: Seaborn python tutorial
Docs: https://seaborn.pydata.org

6. Plotly and Dash [Intermediate]

Explanation (English)

Interactive charts and dashboards in the browser. Install: pip install plotly dash

Tanglish (Tamil + English) Vilakkam

Interactive chart & dashboard - Plotly, Dash.

Example / Program

```
import plotly.express as px
px.scatter(px.data.iris(),x='sepal_width',y='sepal_length',color='species').show()
```

Learn more

YouTube search: Plotly and Dash python tutorial
Docs: https://plotly.com/python/

7. SciPy [Intermediate]

Explanation (English)

Scientific computing: optimisation, integration, statistics, signal processing, sparse matrices, interpolation. Install: pip install scipy

Tanglish (Tamil + English) Vilakkam

SciPy: optimize, stats, sparse, signal.

Example / Program

```
from scipy.optimize import minimize
print(minimize(lambda x:(x[0]-3)**2,[0]).x)
```

Learn more

YouTube search: SciPy python tutorial
Docs: https://docs.scipy.org/doc/scipy/

8. Statsmodels [Intermediate]

Explanation (English)

Classical statistics: OLS/GLM regression with summaries, hypothesis tests, ARIMA/SARIMAX time series. Install: pip install statsmodels

Tanglish (Tamil + English) Vilakkam

Statsmodels: p-value, regression summary, ARIMA.

Example / Program

```
import statsmodels.api as sm,numpy as np
x=sm.add_constant(np.random.rand(50));y=2*x[:,1]+np.random.rand(50)
print(sm.OLS(y,x).fit().summary().tables[1])
```

Learn more

YouTube search: Statsmodels python tutorial
Docs: https://www.statsmodels.org

9. scikit-learn [Beginner]

Explanation (English)

Classical ML: preprocessing, models, pipelines, cross-validation, metrics, model selection. Consistent fit/predict API. Install: pip install scikit-learn

Tanglish (Tamil + English) Vilakkam

scikit-learn = classical ML library. fit(), predict(), pipeline.

Example / Program

```
from sklearn.datasets import load_iris
from sklearn.ensemble import RandomForestClassifier
X,y=load_iris(return_X_y=True)
print(RandomForestClassifier().fit(X,y).score(X,y))
```

Learn more

YouTube search: scikit-learn python tutorial
Docs: https://scikit-learn.org/stable/

10. XGBoost / LightGBM / CatBoost [Intermediate]

Explanation (English)

Gradient boosted trees - state of the art for tabular data and Kaggle. Install: pip install xgboost lightgbm catboost

Tanglish (Tamil + English) Vilakkam

Tabular data-ku best models: XGBoost, LightGBM, CatBoost.

Example / Program

```
import xgboost as xgb
from sklearn.datasets import load_breast_cancer
X,y=load_breast_cancer(return_X_y=True)
print(xgb.XGBClassifier(n_estimators=50).fit(X,y).score(X,y))
```

Learn more

YouTube search: XGBoost / LightGBM / CatBoost python tutorial
Docs: https://xgboost.readthedocs.io

11. imbalanced-learn [Intermediate]

Explanation (English)

Resampling for imbalanced classes: SMOTE, random under/over sampling. Install: pip install imbalanced-learn

Tanglish (Tamil + English) Vilakkam

Imbalanced data-ku SMOTE use pannalam.

Example / Program

```
from imblearn.over_sampling import SMOTE
# Xr,yr=SMOTE().fit_resample(X,y)
```

Learn more

YouTube search: imbalanced-learn python tutorial
Docs: https://imbalanced-learn.org

12. Optuna / Hyperopt / Keras Tuner [Advanced]

Explanation (English)

Automated hyper-parameter optimisation with pruning and visualisation. Install: pip install optuna

Tanglish (Tamil + English) Vilakkam

Hyperparameter tuning automatic-a pannum.

Example / Program

```
import optuna
s=optuna.create_study()
s.optimize(lambda t:(t.suggest_float('x',-5,5)-2)**2,n_trials=20);print(s.best_params)
```

Learn more

YouTube search: Optuna / Hyperopt / Keras Tuner python tutorial
Docs: https://optuna.org

13. PyTorch [Intermediate]

Explanation (English)

Dynamic-graph deep learning framework with autograd, GPU support, and rich ecosystem. Most popular in research. Install: pip install torch torchvision torchaudio

Tanglish (Tamil + English) Vilakkam

PyTorch = research-la romba popular DL framework. Tensor, autograd, nn.Module.

Example / Program

```
import torch
x=torch.randn(3,requires_grad=True);(x**2).sum().backward();print(x.grad)
```

Learn more

YouTube search: PyTorch python tutorial
Docs: https://pytorch.org/docs/

14. TensorFlow and Keras [Intermediate]

Explanation (English)

Google's DL framework; Keras is the high-level API. Strong deployment tools (TF Serving, TFLite, TF.js). Install: pip install tensorflow

Tanglish (Tamil + English) Vilakkam

TensorFlow/Keras - production deployment-ku nalladhu (TFLite, TF.js).

Example / Program

```
import tensorflow as tf
m=tf.keras.Sequential([tf.keras.layers.Dense(1)]);m.compile('sgd','mse');print(m.summary())
```

Learn more

YouTube search: TensorFlow and Keras python tutorial
Docs: https://www.tensorflow.org/api_docs

15. JAX / Flax [Advanced]

Explanation (English)

NumPy-like API with autodiff, jit and vmap on GPU/TPU. Used in cutting-edge research. Install: pip install jax flax

Tanglish (Tamil + English) Vilakkam

JAX = autodiff + jit + GPU/TPU.

Example / Program

```
import jax,jax.numpy as jnp
print(jax.grad(lambda x:x**2)(3.0))
```

Learn more

YouTube search: JAX / Flax python tutorial
Docs: https://jax.readthedocs.io

16. PyTorch Lightning / fastai / Hugging Face Accelerate [Advanced]

Explanation (English)

Higher-level training loops that remove boilerplate and handle multi-GPU/mixed precision. Install: pip install lightning fastai accelerate

Tanglish (Tamil + English) Vilakkam

Training loop boilerplate-a kuraikkum - Lightning, fastai.

Example / Program

```
# import lightning as L ; class M(L.LightningModule): ... ; L.Trainer(max_epochs=3).fit(M(),dl)
```

Learn more

YouTube search: PyTorch Lightning / fastai / Hugging Face Accelerate python tutorial
Docs: https://lightning.ai/docs

17. ONNX and ONNX Runtime [Advanced]

Explanation (English)

Open model format for cross-framework, fast inference. Install: pip install onnx onnxruntime

Tanglish (Tamil + English) Vilakkam

Model-a ONNX-ku convert panni fast inference pannalam.

Example / Program

```
# torch.onnx.export(model,x,'m.onnx')
# import onnxruntime as ort; ort.InferenceSession('m.onnx')
```

Learn more

YouTube search: ONNX and ONNX Runtime python tutorial
Docs: https://onnxruntime.ai/docs/

18. OpenCV [Intermediate]

Explanation (English)

Classic computer vision: reading images/video, filters, edges, contours, feature detection, face detection. Install: pip install opencv-python

Tanglish (Tamil + English) Vilakkam

OpenCV image/video processing library.

Example / Program

```
import cv2
img=cv2.imread('a.jpg');g=cv2.cvtColor(img,cv2.COLOR_BGR2GRAY);e=cv2.Canny(g,100,200)
```

Learn more

YouTube search: OpenCV python tutorial
Docs: https://docs.opencv.org

19. Pillow, scikit-image, Albumentations [Beginner]

Explanation (English)

Image loading, transformations and augmentation pipelines. Install: pip install pillow scikit-image albumentations

Tanglish (Tamil + English) Vilakkam

Image open/resize/augment panna.

Example / Program

```
from PIL import Image
im=Image.open('a.jpg').resize((224,224))
```

Learn more

YouTube search: Pillow, scikit-image, Albumentations python tutorial
Docs: https://pillow.readthedocs.io

20. torchvision / timm / Ultralytics YOLO [Intermediate]

Explanation (English)

Pretrained vision models, datasets, detection and segmentation tools. Install: pip install torchvision timm ultralytics

Tanglish (Tamil + English) Vilakkam

Pretrained vision models (ResNet, ViT, YOLO).

Example / Program

```
from ultralytics import YOLO
m=YOLO('yolov8n.pt');r=m('a.jpg')
```

Learn more

YouTube search: torchvision / timm / Ultralytics YOLO python tutorial
Docs: https://docs.ultralytics.com

21. NLTK [Beginner]

Explanation (English)

Classic NLP toolkit: tokenisation, stemming, POS tagging, corpora, WordNet. Install: pip install nltk

Tanglish (Tamil + English) Vilakkam

NLTK - basic NLP: tokenize, stemming, stopwords.

Example / Program

```
import nltk;nltk.download('punkt')
from nltk.tokenize import word_tokenize
print(word_tokenize('I love AI'))
```

Learn more

YouTube search: NLTK python tutorial
Docs: https://www.nltk.org

22. spaCy [Intermediate]

Explanation (English)

Industrial NLP pipeline: tokenisation, NER, dependency parsing, similarity. Install: pip install spacy

Tanglish (Tamil + English) Vilakkam

spaCy fast NLP pipeline - NER, POS.

Example / Program

```
import spacy;nlp=spacy.load('en_core_web_sm')
print([(e.text,e.label_) for e in nlp('Google is in California').ents])
```

Learn more

YouTube search: spaCy python tutorial
Docs: https://spacy.io

23. Gensim [Intermediate]

Explanation (English)

Topic modelling (LDA), Word2Vec, FastText, document similarity. Install: pip install gensim

Tanglish (Tamil + English) Vilakkam

Word2Vec, LDA topic modelling.

Example / Program

```
from gensim.models import Word2Vec
m=Word2Vec([['ai','ml'],['ml','dl']],min_count=1);print(m.wv['ml'][:3])
```

Learn more

YouTube search: Gensim python tutorial
Docs: https://radimrehurek.com/gensim/

24. Hugging Face Transformers / Datasets / Tokenizers / PEFT / TRL [Intermediate]

Explanation (English)

Thousands of pretrained models (BERT, GPT, T5, Llama) with unified API; fine-tuning with PEFT (LoRA) and TRL. Install: pip install transformers datasets tokenizers peft trl

Tanglish (Tamil + English) Vilakkam

Hugging Face = pretrained models hub. pipeline() oru line-la use pannalam.

Example / Program

```
from transformers import pipeline
print(pipeline('sentiment-analysis')('I love this!'))
```

Learn more

YouTube search: Hugging Face Transformers / Datasets / Tokenizers / PEFT / TRL python tutorial
Docs: https://huggingface.co/docs

25. sentence-transformers [Intermediate]

Explanation (English)
Sentence/document embeddings for semantic search and RAG. Install: pip install sentence-transformers

Tanglish (Tamil + English) Vilakkam
Text embeddings - semantic search, RAG-ku.

Example / Program
```
from sentence_transformers import SentenceTransformer
m=SentenceTransformer('all-MiniLM-L6-v2');print(m.encode(['hi']).shape)
```

Learn more
YouTube search: sentence-transformers python tutorial
Docs: https://www.sbert.net

26. LangChain / LlamaIndex / LangGraph [Intermediate]

Explanation (English)
Frameworks for LLM apps: prompts, chains, retrievers, agents, memory, graph workflows. Install: pip install langchain llama-index langgraph

Tanglish (Tamil + English) Vilakkam
LLM app build panna framework - chains, RAG, agents.

Example / Program
```
# from langchain_core.prompts import ChatPromptTemplate
# prompt=ChatPromptTemplate.from_template('Explain {x}')
```

Learn more
YouTube search: LangChain / LlamaIndex / LangGraph python tutorial
Docs: https://python.langchain.com

27. Anthropic / OpenAI SDKs [Beginner]

Explanation (English)
Official Python clients to call hosted LLM APIs with chat, tools, streaming, vision. Install: pip install anthropic openai

Tanglish (Tamil + English) Vilakkam
LLM APIs call panna official SDK.

Example / Program
```
import anthropic
c=anthropic.Anthropic()  # needs ANTHROPIC_API_KEY
r=c.messages.create(model='claude-sonnet-5-5',max_tokens=100,messages=[{'role':'user','content':'Hi'}]);print(r.content[0
```

Learn more
YouTube search: Anthropic / OpenAI SDKs python tutorial
Docs: https://docs.anthropic.com

28. Vector DBs: FAISS, Chroma, Qdrant, Pinecone, pgvector [Intermediate]

Explanation (English)
Store embeddings and do nearest-neighbour search for RAG. Install: pip install faiss-cpu chromadb qdrant-client

Tanglish (Tamil + English) Vilakkam
Embeddings store panni similarity search pannalam.

Example / Program
```
import faiss,numpy as np
i=faiss.IndexFlatL2(4);i.add(np.random.rand(10,4).astype('f'));print(i.search(np.random.rand(1,4).astype('f'),2))
```

Learn more
YouTube search: Vector DBs: FAISS, Chroma, Qdrant, Pinecone, pgvector python tutorial
Docs: https://faiss.ai

29. Gymnasium and Stable-Baselines3 [Advanced]

Explanation (English)
RL environments and ready algorithms (PPO, DQN, SAC). Install: pip install gymnasium stable-baselines3

Tanglish (Tamil + English) Vilakkam
RL environments & algorithms.

Example / Program
```
import gymnasium as gym
e=gym.make('CartPole-v1');o,_=e.reset();print(o)
```

Learn more
YouTube search: Gymnasium and Stable-Baselines3 python tutorial
Docs: https://gymnasium.farama.org

30. Prophet / sktime / tsfresh / darts [Intermediate]

Explanation (English)
Time-series forecasting and feature extraction. Install: pip install prophet sktime darts

Tanglish (Tamil + English) Vilakkam
Time series forecasting libraries.

Example / Program
```
# from prophet import Prophet; m=Prophet().fit(df); m.predict(m.make_future_dataframe(30))
```

Learn more
YouTube search: Prophet / sktime / tsfresh / darts python tutorial
Docs: https://facebook.github.io/prophet/

31. SHAP / LIME / Captum [Advanced]

Explanation (English)
Explainable AI: feature attributions for models. Install: pip install shap lime captum

Tanglish (Tamil + English) Vilakkam
Model ethu-nala ippadi predict pannuchu-nu explain pannum.

Example / Program
```
import shap
# e=shap.Explainer(model,X);shap.plots.beeswarm(e(X))
```

Learn more
YouTube search: SHAP / LIME / Captum python tutorial
Docs: https://shap.readthedocs.io

32. NetworkX / PyTorch Geometric / DGL [Advanced]

Explanation (English)
Graph analysis and Graph Neural Networks. Install: pip install networkx torch-geometric

Tanglish (Tamil + English) Vilakkam
Graph data & GNN.

Example / Program
```
import networkx as nx
G=nx.karate_club_graph();print(nx.degree_centrality(G)[0])
```

Learn more
YouTube search: NetworkX / PyTorch Geometric / DGL python tutorial
Docs: https://networkx.org

33. Surprise / implicit / LightFM / RecBole [Intermediate]

Explanation (English)
Recommender system libraries. Install: pip install scikit-surprise implicit

Tanglish (Tamil + English) Vilakkam
Recommender systems build panna.

Example / Program

```
# from surprise import SVD,Dataset; algo=SVD().fit(trainset)
```

Learn more

YouTube search: Surprise / implicit / LightFM / RecBole python tutorial
Docs: https://surprise.readthedocs.io

34. Streamlit / Gradio [Beginner]

Explanation (English)

Turn Python scripts into web apps/demos in minutes. Install: pip install streamlit gradio

Tanglish (Tamil + English) Vilakkam

Model demo web app oru nimishathula.

Example / Program

```
# app.py
import streamlit as st
x=st.slider('x',0,10);st.write(x*2)
# run: streamlit run app.py
```

Learn more

YouTube search: Streamlit / Gradio python tutorial
Docs: https://docs.streamlit.io

35. FastAPI / Flask / Uvicorn [Intermediate]

Explanation (English)

Serve models as REST APIs with validation and docs. Install: pip install fastapi uvicorn flask

Tanglish (Tamil + English) Vilakkam

Model-a REST API-a serve panna.

Example / Program

```
from fastapi import FastAPI
app=FastAPI()
@app.get('/pred')
def pred(x:float): return {'y':2*x}
# uvicorn app:app --reload
```

Learn more

YouTube search: FastAPI / Flask / Uvicorn python tutorial
Docs: https://fastapi.tiangolo.com

36. MLflow / Weights & Biases / TensorBoard / DVC [Intermediate]

Explanation (English)

Track experiments, metrics, artefacts, data versions. Install: pip install mlflow wandb tensorboard dvc

Tanglish (Tamil + English) Vilakkam

Experiment tracking & data versioning.

Example / Program

```
import mlflow
with mlflow.start_run():
    mlflow.log_param('lr',.01);mlflow.log_metric('acc',.9)
```

Learn more

YouTube search: MLflow / Weights & Biases / TensorBoard / DVC python tutorial
Docs: https://mlflow.org/docs/latest/

37. Pydantic, python-dotenv, tqdm, joblib, rich [Beginner]

Explanation (English)

Utility libraries: validation, env variables, progress bars, model persistence, pretty logs. Install: pip install pydantic python-dotenv tqdm joblib rich

Tanglish (Tamil + English) Vilakkam

Utility libs: tqdm progress bar, joblib save model.

Example / Program

```
from tqdm import tqdm
import joblib
for i in tqdm(range(3)): pass
joblib.dump({'a':1},'x.pkl')
```

Learn more

YouTube search: Pydantic, python-dotenv, tqdm, joblib, rich python tutorial
Docs: https://docs.pydantic.dev

38. Dask / PySpark / Ray / Modin [Advanced]

Explanation (English)

Distributed and out-of-core computing for data larger than memory. Install: pip install dask pyspark ray

Tanglish (Tamil + English) Vilakkam

Big data distributed processing.

Example / Program

```
import dask.dataframe as dd
# df=dd.read_csv('big*.csv');df.groupby('a').b.mean().compute()
```

Learn more

YouTube search: Dask / PySpark / Ray / Modin python tutorial
Docs: https://docs.dask.org

39. Jupyter, ipywidgets, papermill, nbconvert [Beginner]

Explanation (English)

Interactive notebooks, widgets, parameterised runs and exports. Install: pip install jupyterlab ipywidgets

Tanglish (Tamil + English) Vilakkam

Notebook interactive-a work panna.

Example / Program

```
# jupyter lab   |  jupyter nbconvert --to html nb.ipynb
```

Learn more

YouTube search: Jupyter, ipywidgets, papermill, nbconvert python tutorial
Docs: https://jupyter.org

40. Web scraping: BeautifulSoup, Scrapy, Selenium, Playwright [Intermediate]

Explanation (English)

Collect data from web pages. Install: pip install beautifulsoup4 scrapy selenium playwright

Tanglish (Tamil + English) Vilakkam

Web data collect panna.

Example / Program

```
from bs4 import BeautifulSoup
print(BeautifulSoup('<p>Hi</p>','html.parser').p.text)
```

Learn more

YouTube search: Web scraping: BeautifulSoup, Scrapy, Selenium, Playwright python tutorial
Docs: https://www.crummy.com/software/BeautifulSoup/bs4/doc/

41. Speech & audio: librosa, whisper, torchaudio, SpeechRecognition [Intermediate]

Explanation (English)

Audio features, speech-to-text and TTS. Install: pip install librosa openai-whisper

Tanglish (Tamil + English) Vilakkam

Audio & speech processing.

Example / Program

```
import librosa
# y,sr=librosa.load('a.wav');mfcc=librosa.feature.mfcc(y=y,sr=sr)
```

Learn more
YouTube search: Speech & audio: librosa, whisper, torchaudio, SpeechRecognition python tutorial
Docs: https://librosa.org